Helix Forge Research Benchmark Design Technical Note

HELIX FORGE LAB / p04-full-008004 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded benchmark design evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000055 (research/papers/2026-07-13/helix-forge-research/failure-analysis/tables/helix-forge-research-failure-analysis-thresholds-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	203	0.885	0.878	0.881
slice-02	218	0.867	0.854	0.860
slice-03	244	0.884	0.867	0.875
slice-04	181	0.866	0.880	0.873
slice-05	168	0.883	0.856	0.869